

HAT-P-54b

R-1496

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_200104 · created 2026-08-01T14:17:37+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458852.8021 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.046
Transit depth (Rp/Rs)^2	3.14 +/- 1.64 [%]
Orbital Inclination (inc)	82.32 +/- 2.44
Airmass coefficient 1 (a1)	1.385 +/- 0.05
Airmass coefficient 2 (a2)	-0.31 +/- 0.14
Scatter in the residuals of the lightcurve fit is	3.59 %
Best Comparison Star	#1 - [165, 377]
Optimal Aperture	6.55
Optimal Annulus	18.35
Transit Duration (day)	0.027 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.046 vs 0.1572 (deviation 0.43σ)
Duration fit vs published	0.027 d vs 0.075 d (deviation 1.85σ)
Mid-time O-C	-5.8 min
Depth SNR	3.8
Residual scatter	3.59 %

Light curve

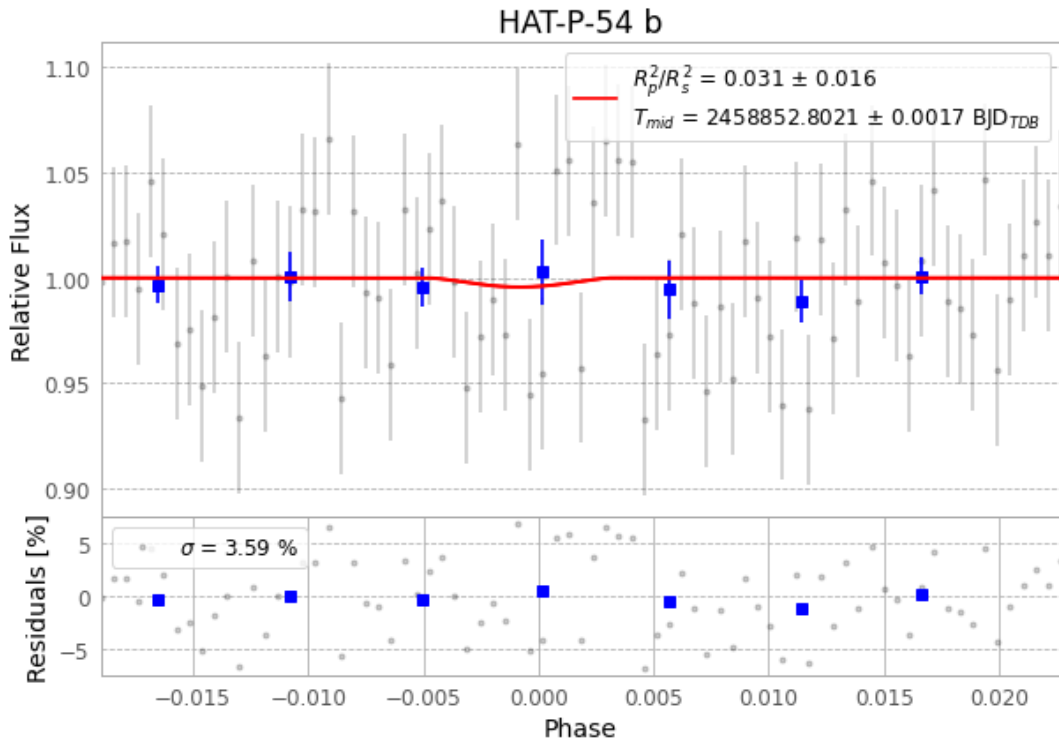


Figure 1 — FinalLightCurve_HAT-P-54 b_2020-01-03.png (SHA-256 8d4ebaee09d414c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [282, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb65564546f7b2af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), HATP-54200104052112.FITS → HATP-54200104090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-200104.log (df6b93b36e40...), FinalLightCurve_HAT-P-54 b_2020-01-03.png (8d4ebaee09d4...), FinalLightCurve_HAT-P-54 b_2020-01-03.csv (9495321a2deb...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:17Z.